

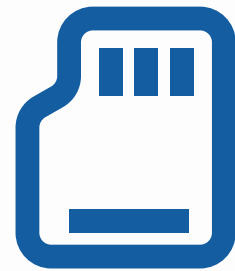
Email Classification System

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Objective:

Build a multi-purpose email classifier



Spam Detection

Identify and filter spam emails



Priority Classification

Assign action priorities
(High, Medium, Low)



Dataset Overview

Source: Gmail API

- Real-time Data Collection: Used Gmail API to fetch live email data from my Gmail account with OAuth 2.0 authentication.
- Diverse Email Types: Retrieved various emails, including spam, personal, and promotional, for multi-task classification (spam and priority).
- Structured Data: Collected email metadata (From, Subject, Date Body) and labels for targeted analysis, stored in excel format for easy processing.
- Efficient API Usage: Managed API rate limits by batching requests and ensuring secure access to the email content.

	From	Subject	Date	Body
0	Kohls@s.kohls.com	Save the date! Get up to 50% off beauty faves ...	2024-11-21 10:44:52-06:00	plus stressfree holiday flawless gift set ...
1	noreply@content.goibibo.com	Bus cancelled = 2x refund!	2024-11-21 19:21:03+05:30	goibibo here's claim offer follow u download g...
2	Kohls@s.kohls.com	Plot twist: Get Black Friday Deals TODAY 🤖 Plu...	2024-11-21 04:11:35-06:00	save 85 clearance mark calendar sephora cyber ...
3	info@members.netflix.com	🔔 Reminder: A Man on the Inside is now on Netflix	2024-11-21 09:56:10+00:00	here reminder start watching rish man inside 2...
4	googlecloud@google.com	Launch and learn with interactive, prebuilt so...	2024-11-21 01:43:02-08:00	google cloud deploy prebuilt google cloud sol...
5	mailers@marketing.goindigo.in	We're celebrating Ethics Week with you	2024-11-21 14:57:48+05:30	share thought feedback honour commitment view ...
6	CaratLane@mailer.caratlane.com	📈 Trending this week: NEW RINGS	2024-11-21 09:04:48+00:00	0
7	noreply@miraeassetmf.co.in	🚀 Launching: Mirae Asset Long Duration Fund	2024-11-21 09:32:18+05:30	click unsubscribe newsletter
8	email@market.temuemail.com	Your Purchase, Our Thanks!	2024-11-21 04:11:54+00:00	temu properly view full message content please...
9	noreply@medium.com	A Founder Who Just Raised a \$3 Million Seed Ro...	2024-11-21 01:50:00+00:00	story rishitha pusapati pusapatirishitha98 ·be...

Data Preprocessing

- ▶ Removing Duplicates
- ▶ Handling Missing Values
- ▶ Removing HTML Tags & URL's
- ▶ Converting to lowercase
- ▶ Removing Punctuation
- ▶ Removing Stopwords
- ▶ Tokenising & Lemmatization

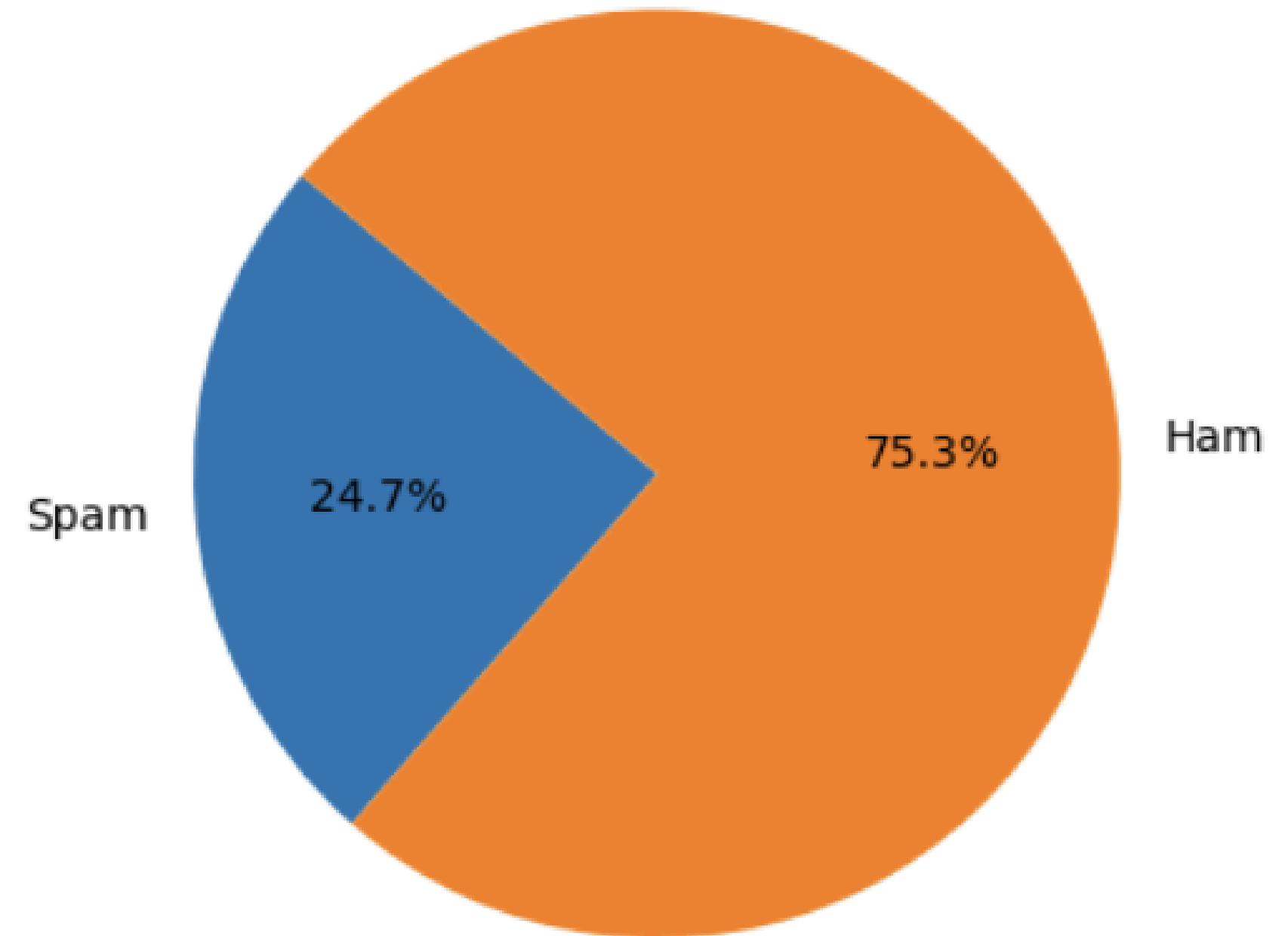


Exploratory Data Analysis (EDA)

Spam-Ham Classification

- Developed a rule-based method to label emails as 'Spam' or 'Ham' using keyword detection.
- Keywords included terms like 'free,' 'urgent,' 'discount,price,buy now,offer' etc.

Initial Data Labeling and Categorization

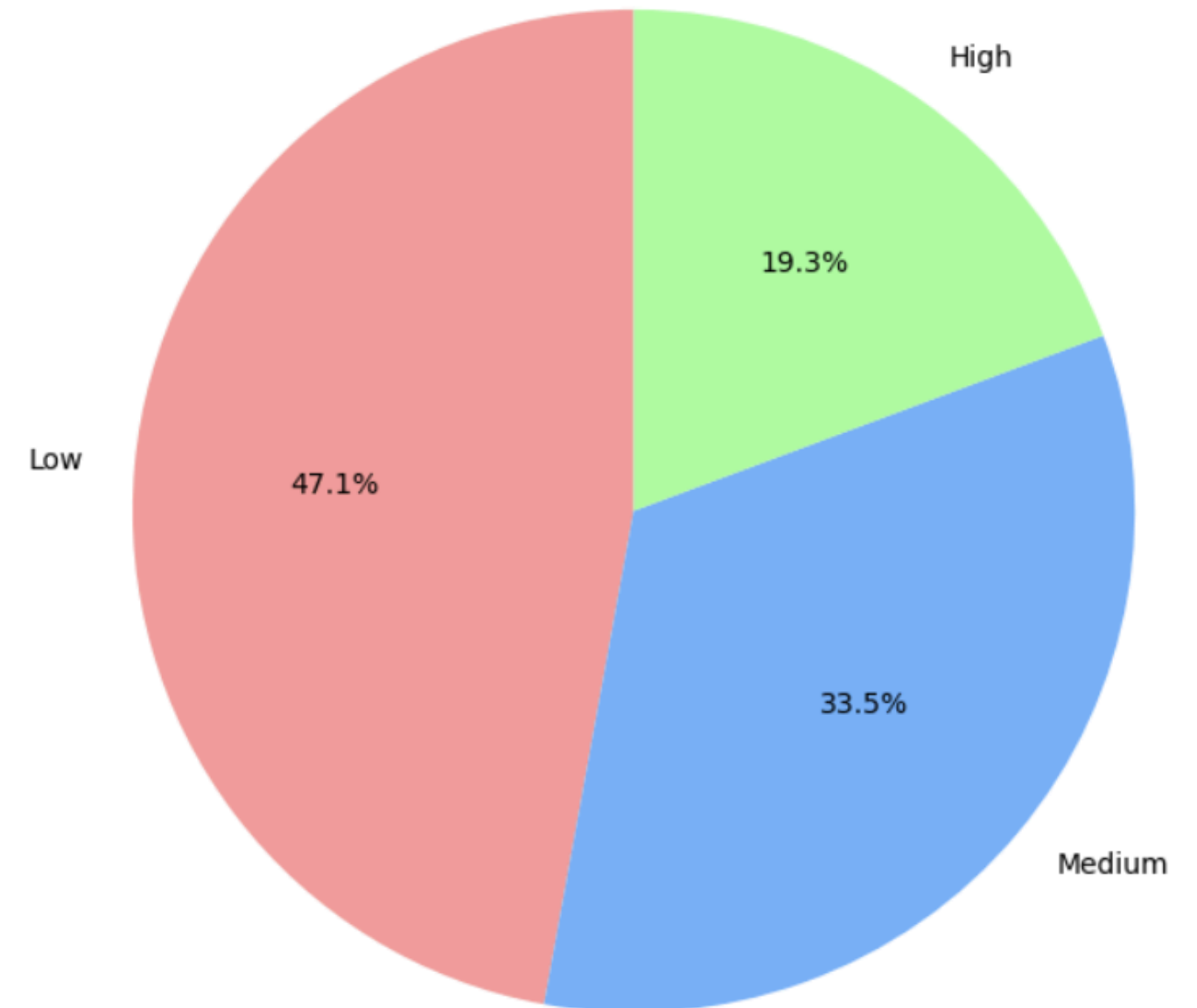


Exploratory Data Analysis (EDA)

Initial Data Labeling and Categorization

Priority Classification

- Classified emails into High, Medium, and Low priority based on priority keywords found in the subject and body of the email.
- Visualized the distribution of email priorities using a pie chart, showing the percentage of emails in each priority category.





SPAM-HAM

Techniques Used

Data Cleaning

- Remove Redundant characters
- Handling Encoded Issues
- Adding Spam feature

Feature Extraction

- Bag-of-Words (BoW)
- TF-IDF
- Word2Vec

Models

- Random Forest Classifier
- Logistic Regression
- XGBoost
- Support Vector Machine (SVM)
- Naive Bayes

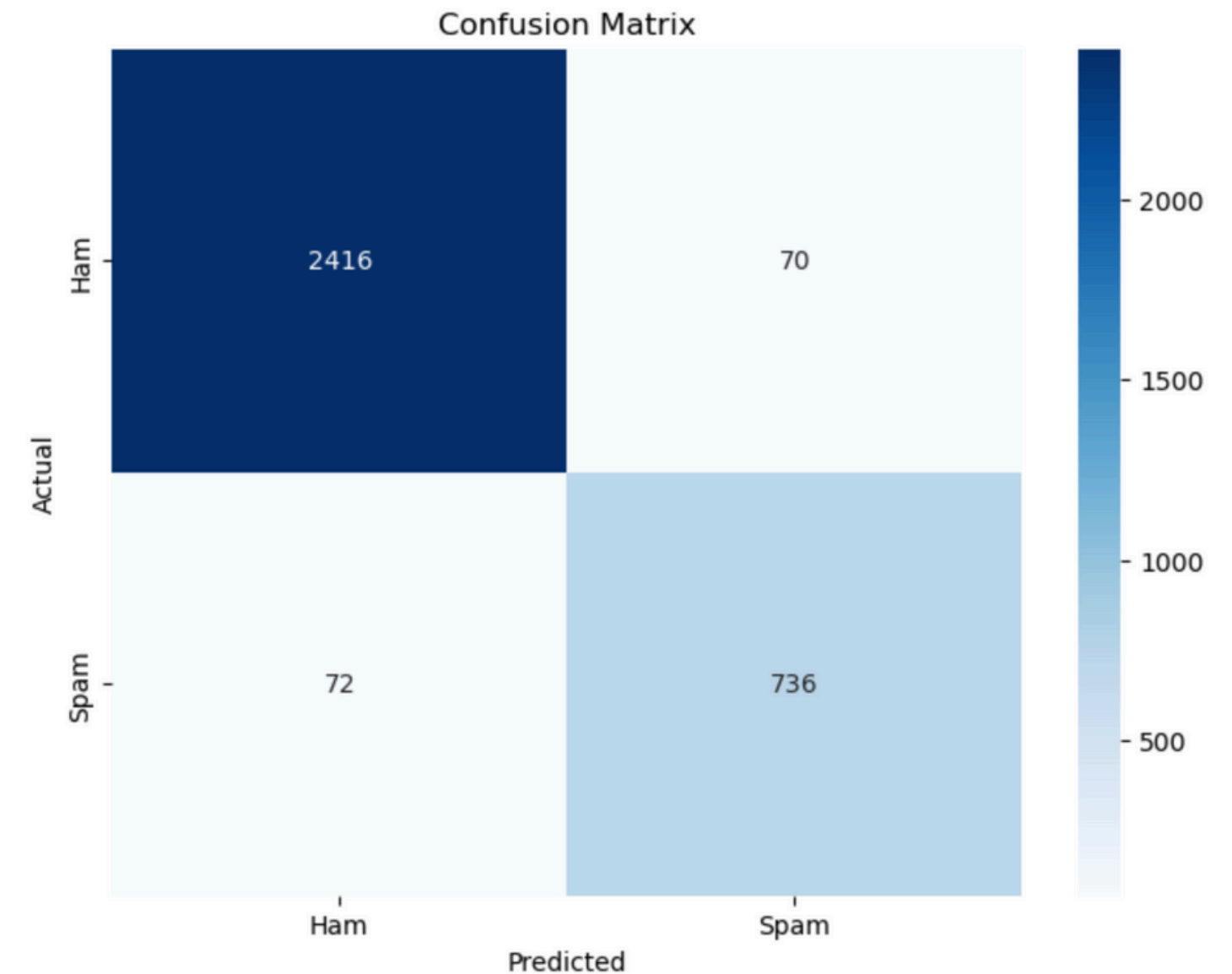
Model Training & Evaluation Results

Model	Feature Extraction	Accuracy	Precision (Spam)	Recall (Spam)	F1-Score (Spam)
Logistic Regression	TF-IDF	0.9827	0.98	0.95	0.96
XGBoost	TF-IDF	0.9809	0.98	0.95	0.96
Random Forest	Bag-of-Words	0.9715	0.99	0.89	0.94
SVM	Bag-of-Words	0.9657	0.93	0.93	0.93
Logistic Regression	Bag-of-Words	0.9702	0.95	0.93	0.94
Naive Bayes	Bag-of-Words	0.9126	0.84	0.79	0.82
Logistic Regression	Word2Vec	0.9065	0.77	0.89	0.82

Confusion Matrix

Performance Metrics:

- True Negatives (Ham correctly classified): 2416.
- True Positives (Spam correctly classified): 736.
- False Negatives (Spam misclassified as Ham): 72.
- False Positives (Ham misclassified as Spam): 70.



**Logistic Regression with TF-IDF was used to generate the confusion matrix.
The model effectively handles imbalanced data and captures spam-related features.**

Why TF-IDF and Logistic Regression?

- XGBoost struggles with noisy and uneven datasets, often favoring the majority class (Ham).
 - Logistic Regression leverages TF-IDF features efficiently and performs better on noisy and imbalanced data.
 - Logistic Regression generalizes rare spam patterns effectively, where XGBoost fails to do so without extensive data cleaning and fine-tuning.
- Logistic Regression provides reliable and consistent results for spam detection, outperforming XGBoost in this scenario.**

TF-IDF and Gradient Boosting (XGBoost)

Email: Hello, I hope you are doing well. Let's catch up soon!	Predicted label: Ham
Email: Limited time offer! Get your free gift card now!	Predicted label: Ham
Email: Are you available for a quick meeting tomorrow? Let me know!	Predicted label: Ham
Email: Congratulations! You've been selected for a special prize!	Predicted label: Ham
Email: Congratulations! You've won a free vacation to Hawaii. Click the link to claim it now.	Predicted label: Ham
Email: Hi John, can we reschedule our meeting for next week?	Predicted label: Ham
Email: Win a brand-new iPhone by signing up here. Limited time offer!	Predicted label: Ham
Email: Reminder: Your subscription is about to expire. Renew it to continue using our service	Predicted label: Ham
Email: The package you ordered will be delivered by tomorrow.	Predicted label: Ham

TF-IDF and Logistic Regression

Email: Hello, I hope you are doing well. Let's catch up soon!	Predicted label: Ham
Email: Limited time offer! Get your free gift card now!	Predicted label: Spam
Email: Are you available for a quick meeting tomorrow? Let me know!	Predicted label: Ham
Email: Congratulations! You've been selected for a special prize!	Predicted label: Spam
Email: Congratulations! You've won a free vacation to Hawaii. Click the link to claim it now.	Predicted label: Spam
Email: Hi John, can we reschedule our meeting for next week?	Predicted label: Ham
Email: Win a brand-new iPhone by signing up here. Limited time offer!	Predicted label: Spam
Email: Reminder: Your subscription is about to expire. Renew it to continue using our service	Predicted label: Ham
Email: The package you ordered will be delivered by tomorrow.	Predicted label: Ham



PRIORITY CLASSIFICATION

Techniques Used

Data preprocessing

- Preparing data for ML models
- Splitting Train and Test Data

Feature Extraction & Hyperparameter

- TF-IDF
- Hyperparameter tuning

Models

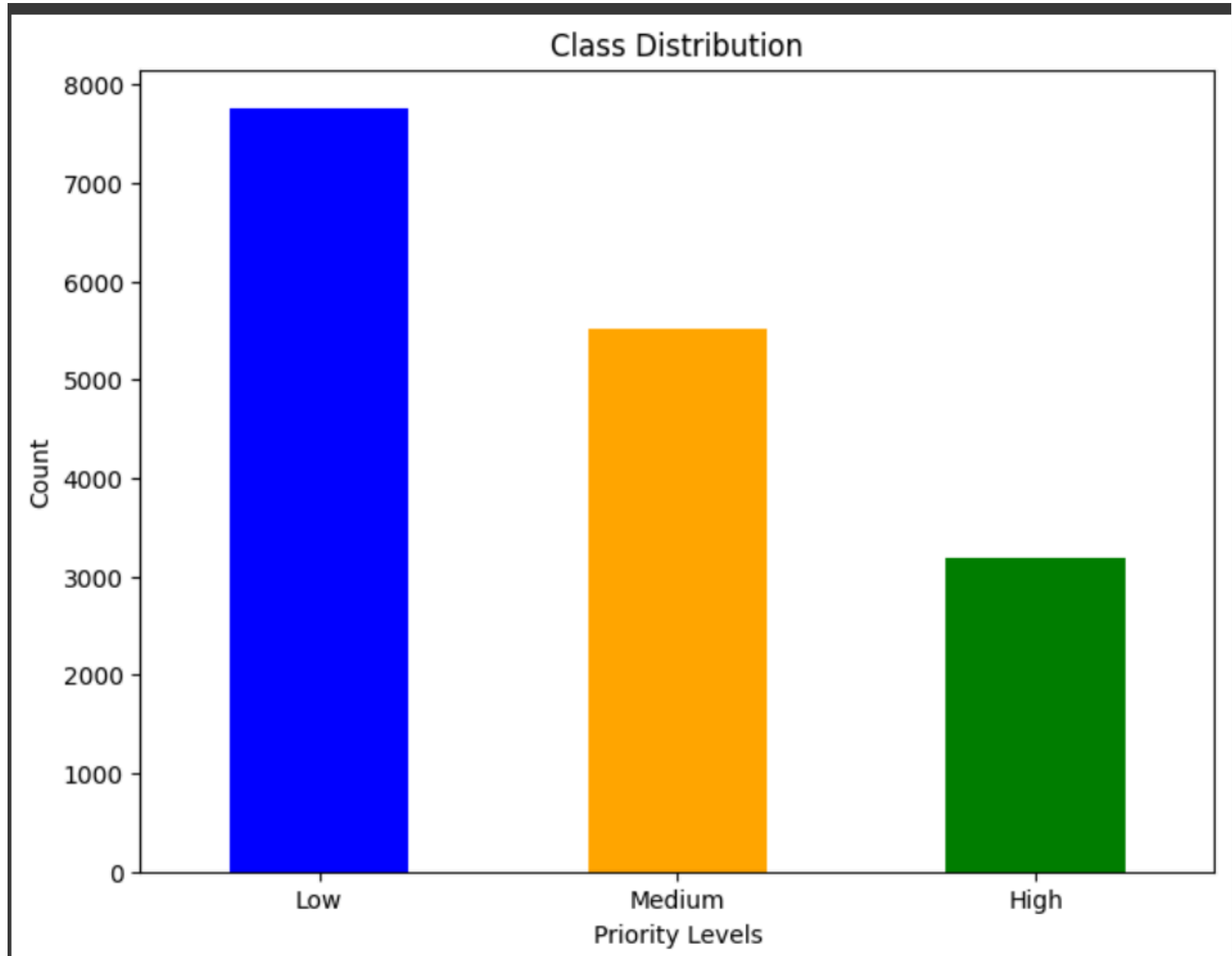
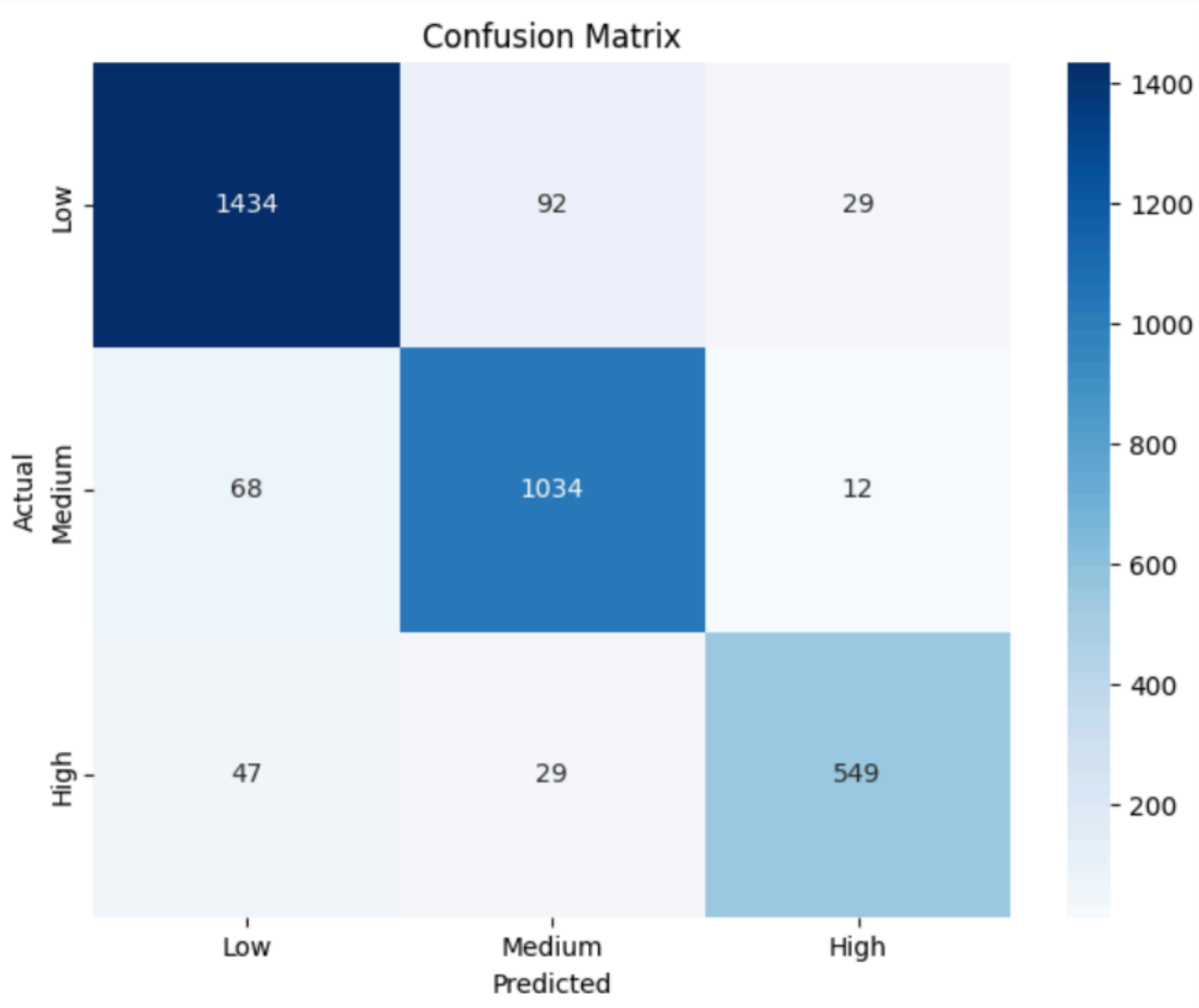
- Random Forest Classifier
- Logistic Regression



Preprocessing Data

- **Merged 'Subject' and 'Body' columns into one feature column, to prepare text data for machine learning model.**

Confusion Matrix and Class Distribution



Confusion matrix shows that the model is good at predicting "Low" and "Medium" priority emails with high accuracy. However, it finds "High" priority emails a bit challenging, leading to some mistakes.



Comparing Both Models

Model	Feature Extraction	Accuracy	Precision (Weighted Avg)	Recall (Weighted Avg)	F1-Score (Weighted Avg)
Logistic Regression	TF-IDF	0.92	0.92	0.92	0.92
Random Forest	TF-IDF	0.95	0.95	0.95	0.95

Comparing results

We tested both models with 10 new examples to predict the priority of each email and compared their performance.

With RF Model

Test Email 1:
Subject: Flash Sale Alert!
Body: Hurry! Get 70% off on electronics. Limited time only.
Predicted Priority: 2

Test Email 2:
Subject: Quarterly Business Review Scheduled
Body: Join us for the QBR scheduled next Monday at 10 AM.
Predicted Priority: 1

Test Email 3:
Subject: Lunch Plans?
Body: Hey, do you want to grab lunch tomorrow? Let me know!
Predicted Priority: 1

Test Email 4:
Subject: Happy Birthday!
Body: Wishing you a fantastic year ahead! Let's celebrate soon.
Predicted Priority: 1

Test Email 5:
Subject: You've Won a \$1,000 Gift Card!
Body: Click here to claim your prize. Offer valid for 24 hours.
Predicted Priority: 0

Test Email 6:
Subject: Action Required: Complete Your Timesheet
Body: Please submit your timesheet by the end of the day.
Predicted Priority: 1

With LR Model

Test Email 1:
Subject: Flash Sale Alert!
Body: Hurry! Get 70% off on electronics. Limited time only.
Predicted Priority: 0

Test Email 2:
Subject: Quarterly Business Review Scheduled
Body: Join us for the QBR scheduled next Monday at 10 AM.
Predicted Priority: 1

Test Email 3:
Subject: Lunch Plans?
Body: Hey, do you want to grab lunch tomorrow? Let me know!
Predicted Priority: 1

Test Email 4:
Subject: Happy Birthday!
Body: Wishing you a fantastic year ahead! Let's celebrate soon.
Predicted Priority: 1

Test Email 5:
Subject: You've Won a \$1,000 Gift Card!
Body: Click here to claim your prize. Offer valid for 24 hours.
Predicted Priority: 0

Test Email 6:
Subject: Action Required: Complete Your Timesheet
Body: Please submit your timesheet by the end of the day.
Predicted Priority: 1



Comparing results

Logistic Regression demonstrated better balance in handling Priority 0(low), 1(medium), and 2(high), making it more versatile and effective for this dataset.

Random Forest Classifier tends to overpredict Priority 1(medium), which could undermine its performance in real-world scenarios.

Even though the Random Forest Classifier demonstrates better precision and results in certain cases, Logistic Regression performs more consistently and delivers better overall performance.

Next Steps...



Local Dialect Support: Adapt replies to regional dialects or localized slang.



Voice-to-Text Integration: For emails with voice notes, convert them to text and reply in the same language.



Adaptive Learning: Continuously train the model to handle nuanced language differences based on user feedback.



THANK YOU!